

UNIVERSAL PLASTIC SHREDDER V2.0

Electrical Control & Power System

Bao Nguyen • Fearghus Tyler • Yaqoub Rabiah • Fox Kang

Portland State University • MCECS • Department of Electrical & Computer Engineering

Sponsor: MME / Ichor Systems (Bridget Lannigan) | Advisors: Prof. Andrew Greenberg, Dr. Jonathan Bird (EE), Robert Paxton (ME)



Maseeh College of Engineering
and Computer Science

PORTLAND STATE UNIVERSITY

The Problem

Ichor Systems generates large volumes of thick rigid plastic (PVC, PP, LDPE) during semiconductor equipment manufacturing. Off-the-shelf shredders are either too small for the material (0.25–0.5 in. sheets, 4–12 in. wide) or priced far above the sponsor's per-line budget. The EE team designed and built the electrical control and power system for an industrial-grade plastic shredder; the mechanical structure is delivered by a separate ME team.

Requirements

Must:

- Shred plastic to ~1 in. pieces.
- Drive a 3–5 HP motor at 70–90 RPM through a controllable VFD.
- Follow NFPA 70 (NEC); include E-stop, overload, and one safety interlock.
- Provide an HMI for operator control and status.
- Stay within the \$1,000 EE-team budget.

Should: status / fault lights, NEMA enclosure, labeled controls.

May: read-only wireless monitoring, data logging, capacitive-touch safety.

System Architecture

The control system is built end-to-end on AutomationDirect industrial hardware – no hobby-grade microcontrollers are in the safety or control path. A PowerTec 71008 disconnect feeds a Mean Well NDR-480-24 24 VDC supply, a Mitsubishi SD-N35 contactor (E-stop hardwired in series with its coil), and a Huanyang HY02D211B-T VFD. The VFD drives a 3 HP bench motor (GE 5KE182BC205B) over U/V/W via blue, white, and red 4 mm banana sockets.



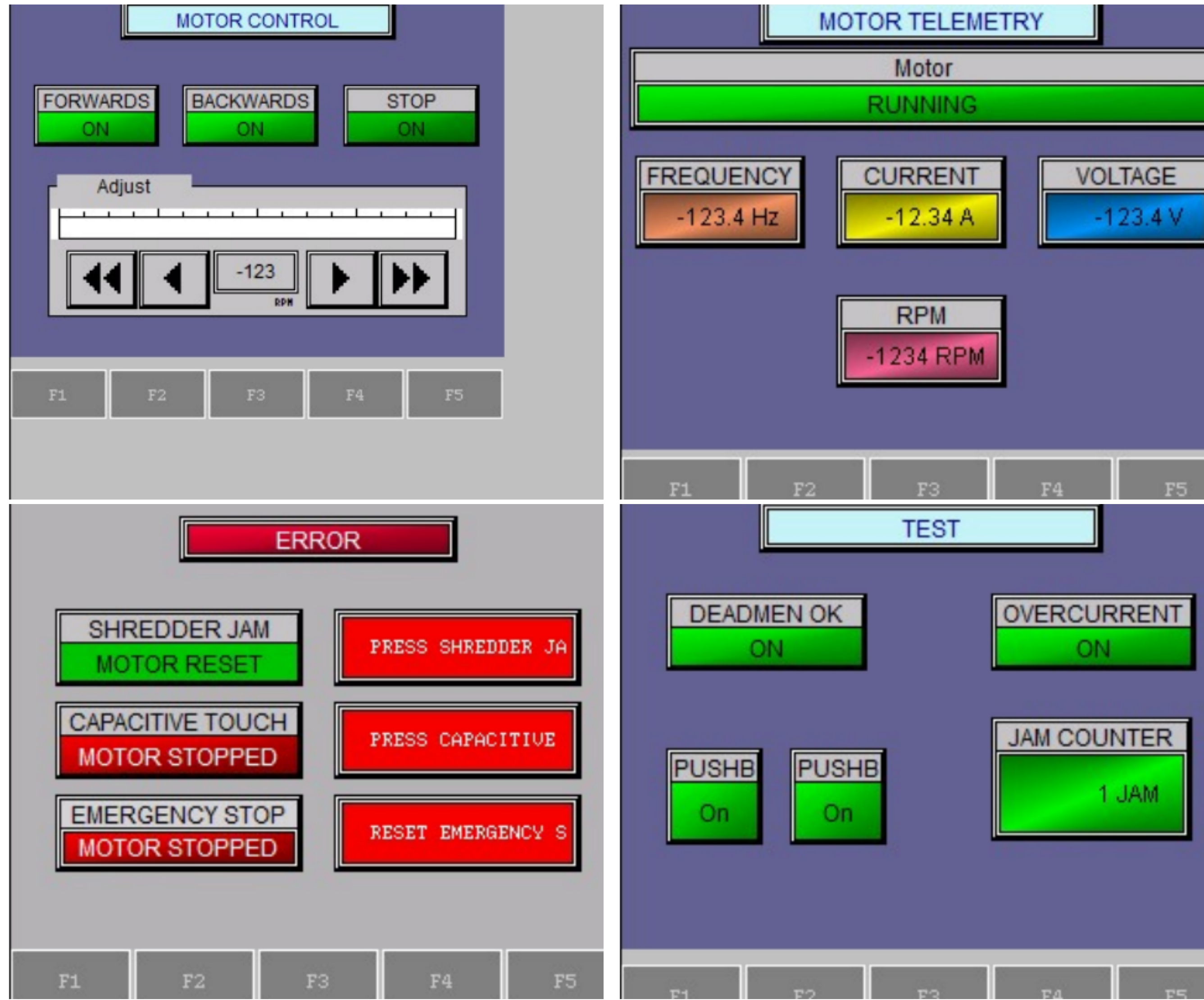
Operator face (left), panel interior (right): HMI, deadman / fwd / rev / E-stop, DirectLOGIC 205, Huanyang VFD, 24 V supply, SD-N35 contactor, terminal blocks.

PLC Control

The PLC is the system controller. A DirectLOGIC 205 D2-09B-1 rack holds the H2-DM1E CPU and five I/O modules: D2-08ND3 (8 DC inputs), F2-08AD-1 (analog current input), F2-04RTD (RTD), D2-08TR (8 relay outputs), and F2-08DA-2 (0–10 V analog output to the VFD). The 31-rung Do-more program implements a deterministic state machine plus a 3-strike jam recovery sequence and a hard-lockout path.

HMI

AutomationDirect EA1-T4CL C-more Micro-Graphic 4 in. touch panel running HMI_413.mgp, serial to the H2-DM1E CPU. Four screens hold 27 PLC tags.



Motor Control, Motor Telemetry, Error / Alarm, Test.

Safety & Compliance

- E-stop:** Category 0 stop. Eaton E22B1 NC mushroom in series with the SD-N35 contactor coil. Drops the contactor independently of the PLC.
- Two-hand / deadman:** both pushbuttons must be held to enable the motor (PLC X2 AND X3).
- Overcurrent / jam:** VFD over-torque trip ($PD124 = 150\%$, $PD125 = 3\text{ s}$) on PLC X5; ladder runs the 3-strike unjam routine.
- Enclosure:** Vorksmen SEB001 steel, NEMA Type 12.
- Compliance:** NFPA 70/79, NEMA 250 / ICS / MG 1, ISO 12100, 13849-1 (Cat B-1 / PL b-c), 13850, 13851, OSHA 1910.147.

Capacitive-Touch Safety (CTSI) Module

A contact-free operator-presence sensor designed and built in-house by Fox Kang. An ESP32 reads a capacitive electrode; on a touch event the ESP32 drives an LED through an optocoupler that closes the 24 V PLC input X7. Power is taken from the panel 24 VDC rail through an LM2596 buck converter to the ESP32's 3.3 V rail.

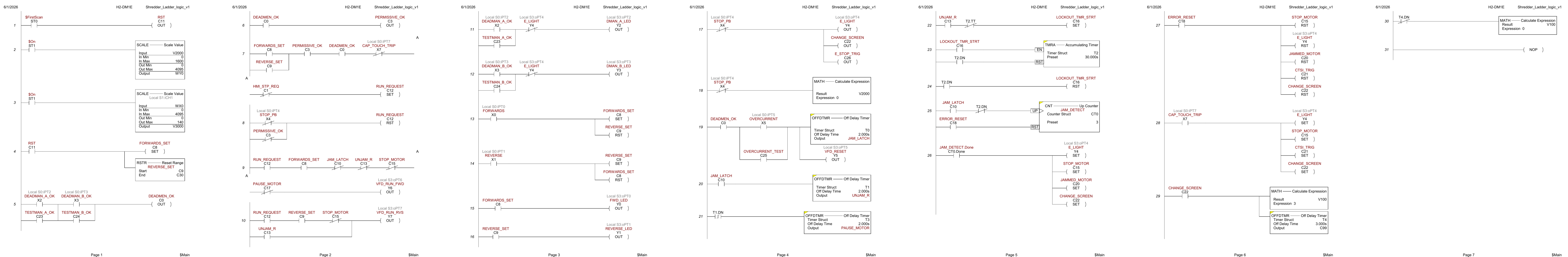
Ladder response (rung 28): touching the electrode latches Y4 (fault LED), C15 (STOP_MOTOR), C21 (CTSI_TRIG), and C22 (CHANGE_SCREEN) – so the motor stops, the panel fault indicator turns on, and the HMI jumps to the ERROR screen simultaneously and independently of the deadman / E-stop chain.

Status: bench-validated against the PLC; full machine-level verification on the enclosed prototype pending.

Results

- Panel built, wired, bench-validated. All 31 rungs program and scan correctly.
- All four HMI screens operate; tag database matches the PLC.
- Speed reference verified: $HMI\ V2000 \rightarrow SCALE \rightarrow F2-08DA-2\ CH1 \rightarrow VFD\ VI$.
- Jam-recovery verified: overcurrent on X5 $\rightarrow 2\text{ s reverse (T1)} \rightarrow$ coast-to-stop buffer (T3).
- Three-revision point list signed off by Fearghus Tyler.

Complete Ladder Logic — 31 rungs across 7 pages



Pages 1–7: power-up init / deadman • RUN.REQUEST gating / Y6 Y7 outputs • indicator LEDs / direction latches • E-stop / jam detection / unjam stroke • lockout timer / jam counter • fault reset / cap-touch / HMI screen change • screen reset / end of program.

Project By the Numbers

- 31 ladder rungs in production.
- 27 HMI tags across 4 screens.
- 14 component sheets in the point list.
- 3 strikes before the motor locks out.
- 2 s unjam reverse stroke.
- 150% over-torque threshold on the VFD.

Hardware Stack

PLC: DirectLOGIC 205 + H2-DM1E.

HMI: C-more EA1-T4CL.

VFD: Huanyang HY02D211B-T.

Motor: GE 5KE182BC205B (3 HP bench).

PSU: Mean Well NDR-480-24.

Contactor: Mitsubishi SD-N35.

Future Work

Production motor selection (3–5 HP) supersedes PD141--PD144.

Mechanical integration and shred-load run pending ME team.

Capacitive-touch (CTSI) module on X7: machine-level verification pending.

Optional Modbus link to the VFD for richer telemetry.

Acknowledgments

Thanks to **Bridget Lannigan** (MME / Ichor Systems) for sponsoring the project; to advisors **Prof. Andrew Greenberg**, **Dr. Jonathan Bird**, and **Robert Paxton**; and to the **PSU Power Engineering Laboratory** for bench access.

ECE 412/413 Capstone Poster Session • PSU MCECS • June 12, 2026